

UGED1111 Logic

Symbolic Language
and
Truth Table

Symbolic language



- To show the logical structure of an argument
- To help checking the validity of an argument

Propositional Logic

- Propositions (/statements)

- $P, Q, R, S, \dots$

- Logical operators(/connectives)

- $\sim, \vee, \cdot, \supset, \equiv$

Truth Function



A truth function is a function that takes truth values as input and generate a truth value as output.

Propositional Logic is a truth functional system.

The truth function is any compound proposition whose truth value is completely determined by the truth values of its components.

Simple statement

A simple statement /atomic statement is one that does not contain any other statement as a component.

Examples:

1. Human activity is the main reason of global warming.
2. 教育局禁止學生會舉辦丘比特活動。
3. 我在NA CAN 吃午餐。
4. 中華白海豚的棲息地不停被破壞。
5. Judy attends today's class.

Compound statement

A compound statement (複合句) is one that contains at least one simple statement as a component.

Example:

1. **It is not the case that** human activity is the main reason of global warming.
2. 明愛禁止學生會舉辦丘比特活動，而且這事被廣泛報導。
3. 我在NA CAN 吃午餐，**或者**我在MED CAN 吃午餐。
4. **如果**香港繼續填海，則中華白海豚的棲息地不停被破壞。
5. Judy will attend today's class **if and only if** Tiffany will attend today's class.

Operators

Logical operators: $\sim, \vee, \cdot, \supset, \equiv$

Operators you have used in daily life: $+, -, \times, \div$

➤ $(x + y), (x - y), (x \times y), (x \div y)$

□ $\sim(P \equiv (Q \vee R)) \supset (\sim R \cdot S)$

□ $[9 + (2 - 1) \times 6] \div 3$

Logical Operators

Operator	Name	Logical function	Used to translate
$\sim$	Tilde	Negation (否定句)	Not, it is not the case that...
			不是，不，並非...
$\cdot$	Dot	Conjunction (合取句)	And, also, but...
			及，而且，但是...
$\vee$	Wedge	Disjunction (分取句)	...or...
			或
$\supset$	Horseshoe	Conditional (條件句)	If...then, only if...
			如果...則，僅當...
$\equiv$	Triple bar	Bi-conditional (雙條件句)	If and only if
			當且僅當

Negation

To translate “not”.

Example:

Snow is white. (**P**)

Snow is not white. ($\sim$ **P**)

p	$\sim$p
T	F
F	T

Negation



What is the negation of this statement?

“Bella likes expensive dresses.”

To properly express its negation:

“It is false that Bella likes expensive dresses.”

“It is not the case that Bella likes expensive dress.”

Conjunction

To translate “and”

Example:

Both *Ironman* and *Ant-man* are
Marvel Characters. ($P \cdot Q$)

p	q	$p \cdot q$
T	T	T
T	F	F
F	T	F
F	F	F

- *Ironman* is a Marvel Character. (P)
- *Ant-man* is a Marvel Character. (Q)

Disjunction

To translate “or”

我上課或者我睡覺。 ($P \vee Q$)

- 我上課 (P) ; 我睡覺 (Q)

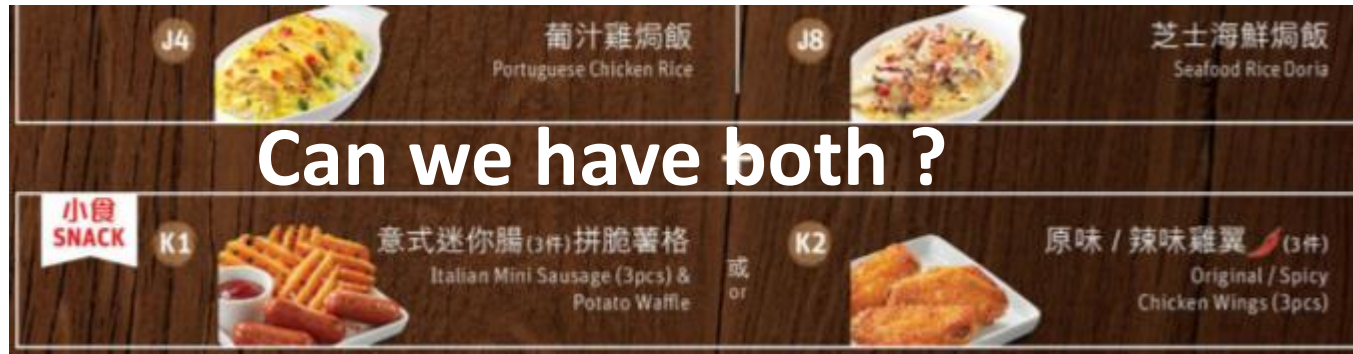
Either you are clever or you will fail the course.

($M \vee N$) ?

- You are clever. (M)
- You will fail the course. (N)

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Disjunction



Example :You're either with us, or against us.

Inclusive-or	Exclusive-or
$p \vee q$	$(p \vee q) \cdot \sim(p \cdot q)$

Conditional

To translate “if...then”

If Joe feels bored, he will read books about logic. ($P \supset Q$)

- Joe feels bored. (P)
- Joe will read books about logic. (Q)

p	q	$p \supset q$
T	T	T
T	F	F
F	T	T
F	F	T

Conditional

$P \supset Q$

If P, then Q

“P” is antecedent (前項/前件).

“Q” is consequent (後項/後件).

Material conditional

The whole conditional is false when and only when the antecedent is true and the consequent is false.

Conditional

Counter-intuitive ?

Why the conditional is true if the antecedent is false?

It seems that we won't accept “如果地球是正方體，則比卡超存在” is true.

p	q	$p \supset q$
T	T	T
T	F	F
F	T	T
F	F	T

Conditional

Reasons:

1. Ensure that the truth-value of " $(\phi \supset \phi)$ " is always T whether ϕ is T or F.

E.g 如果美斯是女人，則美斯是女人。

p	q	$p \supset q$
T	T	T
T	F	F
F	T	T
F	F	T

Conditional

Any cases that we endorse that a conditional is true if the antecedent is false ?

Example given by Copi

Given $2 < 4$. From this, we know that all number which is small than 2 will be smaller than 4,

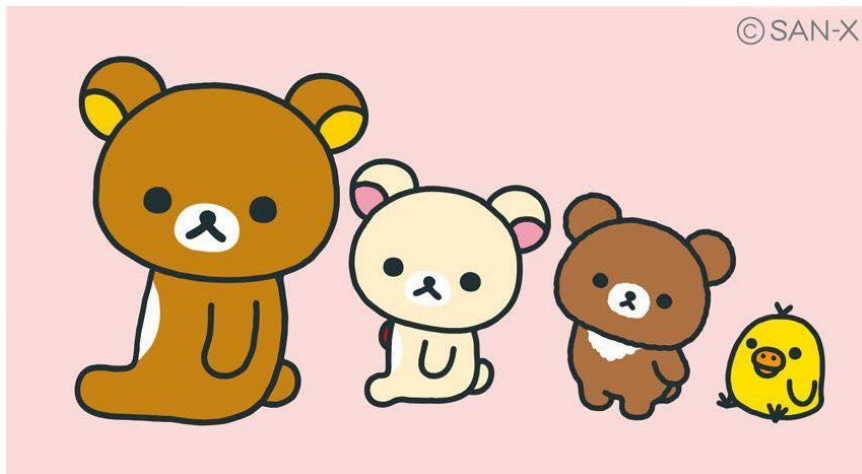
If $x < 2$, then $x < 4$.

Therefore,

If $3 < 2$, then $3 < 4$.

If $5 < 2$, then $5 < 4$.

A Further Example



鬆弛熊高過蜜茶熊. 由此得知任何高過鬆弛熊的生物都比蜜茶熊高

If A 比鬆弛熊高, then A比蜜茶熊高.

所以,

If白熊比鬆弛熊高, then白熊比蜜茶熊高。

If小雞比鬆弛熊高, then小雞比蜜茶熊高

Bi-conditional

To translate “if and only if”
(iff)

June gets an A iff she gets full-
mark. ($P \equiv Q$)

- June gets an A. (P)
- June gets full-mark. (Q)

p	q	$p \equiv q$
T	T	T
T	F	F
F	T	F
F	F	T

Bi-conditional



It means $(P \supset Q) \cdot (Q \supset P)$.

The whole bi-conditional is true when and only when the antecedent and the consequent are of the **same truth value**.

Well-formed Formulas(Wffs)

Wff: Syntactically correct arrangement of symbols

Rules for Well-formed Formulas :

1. Capital letters are wffs.
2. If α is a wff, then so is $\sim\alpha$.
3. If α and β are wffs, then so is $(\alpha\cdot\beta)$.
4. If α and β are wffs, then so is $(\alpha\vee\beta)$.
5. If α and β are wffs, then so is $(\alpha\supset\beta)$.
6. If α and β are wffs, then so is $(\alpha\equiv\beta)$.
7. Nothing else counts as a wff.

Operators

Consider $x \times y$, suppose $x = 1$ or 0 and $y = 1$ or 0

x	y	$x \times y$
1	1	1
1	0	0
0	1	0
0	0	0

p	q	$p \cdot q$
T	T	T
T	F	F
F	T	F
F	F	F

you may also substitute **(a + b) into x**

Operators

Consider $(x + y) \times y$,

suppose $x = 1$ or 0 and $y = 1$ or 0

x	y	$(x + y)$	$\times$	y
1	1	2	2	1
1	0	1	0	0
0	1	1	1	1
0	0	0	0	0

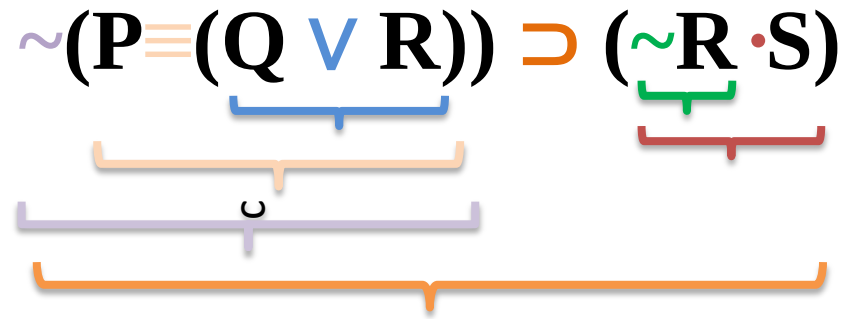
P	Q	$(P \supset Q)$	$\cdot$	Q
T	T	T	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	F	F

Operators and Brackets

- In a compound statement, it may includes more than one logical operator.
 - ◆ $\sim(P \equiv (Q \vee R)) \supset (\sim R \cdot S)$
- Brackets are used to **avoid ambiguity**.
- Among those operators in the formulas, **one of them is the main operator**, and the rest are minor operators.

Operators and Brackets

- The main operator is the operator that has everything else in the statement as its scope.
- The minor operators are the operators that have only the component of the whole statement as its scope.



Operators and Brackets

□ An unary operator is an operator that operates on exactly one wff.

□ A binary operator is an operator that operates on exactly two wffs.

Unary	Binary
$\sim$	$\cdot, \vee, \supset, \equiv$
$\sim\alpha$	$\alpha \cdot \beta$ $\alpha \vee \beta$ $\alpha \supset \beta$ $\alpha \equiv \beta$

Discussion 1

Which of the following formulas are wffs?

1. $\sim\sim\sim P \vee R$

2. $\sim PQ \vee R$

3. $A \subset B$

4. $A \sim B$

5. $p \vee q$

6. $P \supset \sim R$

7. Z

8. $\supset \sim R \cdot S$

9. $M \equiv N \equiv Q$

10. $((P \supset R) \cdot (S \vee R)) \equiv S$

11. $P \sim \supset R$

12. $M \leftrightarrow Q$

Truth Table

Truth table: An arrangement of truth values that shows in every possible case how the truth value of a compound proposition is determined by the truth values of its simple components.

One row = one logical possibility

P	Q	$P \supset Q$
T	T	T
T	F	F
F	T	T
F	F	T

Truth Table

To construct a complete truth table,

1. Enter the truth values of the simple components.
2. Compute the truth values of the compound components.

**** we have to start from the components in the innermost brackets.**

We have to determine the **number of rows** of it first .
But how?

Procedures

The number of rows = 2^n , where n is the number of different simple proposition components.

The next step is to divide the number of lines in half. Then *we assign T to the half of the lines ($2^n/2$ lines) for each letter and F on the remaining lines.*

Circle the column of main operator

****Make sure every logical combination is assigned to the truth table**

Example 1

Construct a truth table for:

$$\sim(P \cdot (Q \equiv \sim P))$$

P	Q	$\sim (P \cdot (Q \equiv \sim P))$

Example 1

Construct a truth table for:

$$\sim(P \cdot (Q \equiv \sim P))$$

P	Q	$\sim$	(P	$\cdot$	(Q	$\equiv$	$\sim$	P))
T	T	T	T	F	T	F	F	T
T	F	F	T	T	F	T	F	T
F	T	T	F	F	T	T	T	F
F	F	T	F	F	F	F	T	F

Discussion 1

Construct a truth table for:

$$\sim P \vee (\sim Q \supset P)$$

P	Q	$\sim P \vee (\sim Q \supset P)$
T	T	
T	F	
F	T	
F	F	

Discussion 1

P	Q	$\sim$	P	$\vee$	$(\sim$	Q	$\supset$	P)
T	T	F	T	T	F	T	T	T
T	F	F	T	T	T	F	T	T
F	T	T	F	T	F	T	T	F
F	F	T	F	T	T	F	F	F